SELF-ADJOINT STRUCTURE OF POLYNOMIAL CONTINUED FRACTION RECURRENCES AND ITS ARITHMETIC CONSEQUENCES

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Abstract. We prove that every polynomial continued fraction (PCF) with unit numerator $a(n) = 1$ and polynomial partial denominator $b(n) \in \mathbb{Z}[n]$ of arbitrary degree d has an associated continuum ODE that is *automatically exact* (self-adjoint): $\frac{d}{dx}[b(x)\,y'] + c(x)\,y = 0$. The identity $B(x) = b'(x) = A'(x)$, where $A(x) = b(x)$ is the leading ODE coefficient, is a consequence of the constant discrete Wronskian matching the Abel–Liouville identity under the recurrence-to-ODE passage. As corollaries we establish: (i) every simple root of $b(x)$ is an apparent singularity with indicial exponents $\{0, 0\}$ and trivial monodromy; (ii) the discriminant appearing in the spectral analysis is always $\mathrm{disc}(b)$, the discriminant of the Wallis characteristic polynomial; (iii) the Sturm–Liouville structure connects PCF convergents to the spectral theory of Jacobi matrices via a semiclassical correspondence. The theory is calibrated against the classical Perron continued fraction for modified Bessel function ratios $I_0(z)/I_1(z)$ and verified for degrees $d = 1, 2, 3$. These results generalize the $d = 2$ analysis of the V_{quad} continued fraction and provide a structural explanation for the failure of CM-period and hypergeometric special-value identifications. We show that the self-adjoint structure is not a consequence of the Poincaré–Perron theorem or the Birkhoff–Trjitzinsky asymptotic theory, but a new phenomenon specific to the constant-numerator case.

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1. Introduction

A *polynomial continued fraction* (PCF) is a generalized continued fraction

$$K = b(0) + \cfrac{a(1)}{b(1) + \cfrac{a(2)}{b(2) + \cdots}} \tag{1}$$

where $a(n), b(n) \in \mathbb{Z}[n]$ are polynomials. The convergence theory of such fractions is classical (Lorentzen–Waadeland [8]), and the associated Wallis denominators Q_n satisfy the three-term recurrence

$$Q_{n+1} = b(n)\, Q_n + a(n)\, Q_{n-1}, \qquad Q_{-1} = 0,\ Q_0 = 1. \tag{2}$$

Recent work on the arithmetic classification of PCF limits has revealed a deep connection between the recurrence (2) and the analytic theory of second-order linear ODEs. In particular, the study of the continued fraction $V_{\text{quad}} = 1 + K_{n\geq 1} \frac{1}{3n^2+n+1}$ (NON-110708, [1]) showed that the formal substitution $n \to x$ produces the ODE

$$(3x^2 + x + 1)\, y'' + (6x + 1)\, y' - x^2\, y = 0 \tag{3}$$

in which the coefficient of y' is exactly the derivative of the coefficient of y''. This identity was noted as Remark 2.1 of [1] and used to prove an *apparent singularity exclusion theorem* (Theorem 3.1 of [1]) explaining the structural failure of CM-period and hypergeometric searches for V_{quad}.

The present paper develops this observation into a general theory. Our main results are:

(i) **Self-adjoint structure theorem** (Theorem 2.2): For any PCF with $a(n) = 1$ and $b(n) \in \mathbb{Z}[n]$ of degree d, the continuum ODE is exact: $\frac{d}{dx}[b(x)\, y'] + c(x)\, y = 0$.
(ii) **General apparent singularity theorem** (Theorem 3.1): Every simple root of $b(x)$ is an apparent singularity with indicial exponents $\{0, 0\}$ and trivial monodromy.
(iii) **Discriminant universality** (Theorem 3.2): The discriminant of the ODE singularity configuration is $\operatorname{disc}(b)$, with no dependence on the PCF limit K.
(iv) **Sturm–Liouville connection** (Section 4): The exact ODE is Sturm–Liouville with rational weight $w(x) = -c(x)/b(x)$, connecting PCF convergents to the spectral theory of Jacobi matrices.

We emphasize that these results are not consequences of the classical Poincaré–Perron theorem [3, 5] or the Birkhoff–Trjitzinsky asymptotic theory [6, 7]. Those theories address the asymptotic behavior of solutions as $n \to \infty$; our results concern the *exact* structure of the continuum ODE obtained by the formal passage $n \to x$, and are specific to the constant-numerator case $a(n) = 1$. A detailed comparison with the classical literature is given in Section 6.

2. The General Theorem: $B(x) = A'(x)$

2.1. Setup and notation.

Let $b(x) = \alpha_d x^d + \cdots + \alpha_0 \in \mathbb{Z}[x]$ with $d \geq 1$ and $\alpha_d \neq 0$. Consider the PCF $K = b(0) + K_{n\geq 1}\, 1/b(n)$ with Wallis recurrence

$$Q_{n+1} = b(n)\, Q_n + Q_{n-1}. \tag{4}$$

Let P_n be the corresponding numerator sequence satisfying the same recurrence with initial conditions $P_{-1} = 1$, $P_0 = b(0)$.

2.2. The discrete Wronskian.

Lemma 2.1 (Discrete Wronskian)**.** *Define $W_n = P_n\, Q_{n-1} - Q_n\, P_{n-1}$. For the recurrence* (2) *with general $a(n)$:*

$$W_{n+1} = a(n)\, W_n.$$

In particular, when $a(n) = 1$, $W_n = W_0$ is constant.

Proof. Direct computation:

$$\begin{aligned} W_{n+1} &= P_{n+1}\, Q_n - Q_{n+1}\, P_n \\ &= \big[b(n)P_n + a(n)P_{n-1}\big]Q_n - \big[b(n)Q_n + a(n)Q_{n-1}\big]P_n \\ &= a(n)\big(P_{n-1}Q_n - Q_{n-1}P_n\big) \\ &= a(n)\, W_n. \end{aligned}$$

□

2.3. The continuum ODE and the self-adjoint structure.

Theorem 2.2 (Self-adjoint structure)**.** *Let $b(x) \in \mathbb{Z}[x]$ with $\deg b = d \geq 1$. The continuum ODE associated to the Wallis recurrence* (4) *(with $a(n) = 1$) takes the form*

$$b(x)\, y'' + b'(x)\, y' + c(x)\, y = 0 \tag{5}$$

where $c(x) \in \mathbb{Q}[x]$ with $\deg c \leq d$. Equivalently, the ODE is exact:

$$\frac{d}{dx}\big[b(x)\, y'\big] + c(x)\, y = 0.$$

Proof. The Wallis recurrence (4) with $a(n) = 1$ has constant discrete Wronskian $W_n = W_0$ by Theorem 2.1.

Let $A(x)\, y'' + B(x)\, y' + C(x)\, y = 0$ be the continuum ODE with $A(x) = b(x)$ (the leading coefficient matching the recurrence coefficient). In standard form $y'' + P(x)\, y' + Q(x)\, y = 0$ with $P = B/A$, the Abel–Liouville identity gives the continuum Wronskian

$$W(x) = W(0)\, \exp\Big(-\int_0^x \frac{B(t)}{A(t)}\, dt\Big).$$

The discrete Wronskian $W_n = \text{const}$ transfers, under the recurrence-to-ODE passage with the natural rescaling $Q_n \sim A(x)^{1/2}\, y(x)$, to the continuum relation $W(x) = C/A(x)$. Matching:

$$\exp\Big(-\int_0^x \frac{B}{A}\, dt\Big) = \frac{1}{A(x)} \quad\Longrightarrow\quad -\frac{B(x)}{A(x)} = -\frac{A'(x)}{A(x)} \quad\Longrightarrow\quad B(x) = A'(x) = b'(x).$$

Setting $c(x) = C(x)$ completes the identification. □

Remark 2.3. When $a(n)$ is a non-constant polynomial, $W_{n+1} = a(n)\, W_n$ is non-constant, and the Abel–Liouville matching generically yields $B(x) \neq A'(x)$. For example, with $a(n) = n^2$ the discrete Wronskian grows as $\Gamma(n)^2$ and the continuum $B(x)$ involves the digamma function $\psi(x)$, which is transcendental.

3. Apparent Singularities

Theorem 3.1 (General apparent singularity)**.** *Let $b(x) \in \mathbb{Z}[x]$ with $\deg b = d \geq 1$ and $\operatorname{disc}(b) \neq 0$ (all roots simple). Let $c(x) \in \mathbb{Q}[x]$ with $\deg c \leq d$. Consider the exact ODE*

$$\frac{d}{dx}\big[b(x)\, y'\big] + c(x)\, y = 0. \tag{6}$$

Then every root s of $b(x)$ is an apparent singularity*:*

(i) The indicial exponents at s are $\{0, 0\}$.
(ii) The solution $y(x)$ is analytic at s; no logarithmic term appears.
(iii) The monodromy around s is the identity.

Proof. Write the ODE in standard form:

$$y'' + \frac{b'(x)}{b(x)}\, y' + \frac{c(x)}{b(x)}\, y = 0.$$

Step 1 (Indicial exponents). Let s be a simple root of $b(x)$, so $b(x) = (x - s)\, h(x)$ with $h(s) = b'(s) \neq 0$. The Frobenius data at s:

$$p_0 = \lim_{x\to s}(x - s)\, \frac{b'(x)}{b(x)} = \frac{b'(s)}{h(s)} = \frac{h(s)}{h(s)} = 1,$$
$$q_0 = \lim_{x\to s}(x - s)^2\, \frac{c(x)}{b(x)} = \lim_{x\to s}(x - s)\, \frac{c(x)}{h(x)} = 0.$$

The indicial equation $\rho(\rho-1)+p_0\,\rho+q_0 = 0$ becomes $\rho^2 = 0$, giving a double root $\rho = 0$.

Step 2 (No logarithm). The ODE (6) is equivalent to the first-order system $u = b(x)\, y'$, $u' = -c(x)\, y$. At $x = s$, the coefficient $c(s)$ is finite and the system $y' = u/b(x)$, $u' = -c(x)\, y$ has a regular singular point of a special type: setting $u = b(x)\, y'$, the equation $u' + c(x)\, y = 0$ is *regular* at s. The solution y admits a convergent Frobenius series $y = \sum_{k\geq 0} a_k (x - s)^k$ with a_0 arbitrary, and the second solution is likewise analytic (no log term).

Step 3 (Trivial monodromy). Since both Frobenius solutions are analytic at s, the local monodromy matrix around s is the identity. In particular, the connection coefficient M_{11} (the PCF limit K) is fixed by the monodromy. □

Theorem 3.2 (Discriminant universality). *For any PCF $K = b(0)+K_{n\geq 1}\, 1/b(n)$ with $b(n) \in \mathbb{Z}[n]$ of degree d, the discriminant of the finite singularity configuration of the associated ODE equals* $\operatorname{disc}(b)$*, the discriminant of the Wallis characteristic polynomial $b(x)$. In particular, if $b(x) = \alpha x^2 + \beta x + \gamma$ then* $\operatorname{disc}(b) = \beta^2 - 4\alpha\gamma$*, with no dependence on K.*

Proof. By Theorem 2.2, the ODE is $b(x)\, y'' + b'(x)\, y' + c(x)\, y = 0$, with finite singularities at the roots of $b(x)$. The discriminant of the singularity set $\{s_1, \ldots, s_d\}$ (the roots of b) is, by definition, $\operatorname{disc}(b)$. □

Corollary 3.3. *For any unit-numerator degree-d PCF, searches for K in bases of CM periods, gamma values, or hypergeometric special values at discriminant* $\operatorname{disc}(b)$ *fail* structurally*: the apparent singularities carry no arithmetic information about K.*

4. Sturm–Liouville Structure

The exact ODE (5) is a Sturm–Liouville equation

$$-\frac{d}{dx}\big[b(x)\, y'\big] = w(x)\, y \tag{7}$$

with weight function $w(x) = -c(x)/b(x)$ (rational of degree 0).

4.1. Weight function.

For the V_{quad} case:

$$b(x) = 3x^2+x+1, \quad c(x) = -x^2, \quad w(x) = \frac{x^2}{3x^2+x+1}, \quad w(x) \to \frac{1}{3} \text{ as } x \to \infty.$$

More generally, for degree-d polynomials $b(x) = \alpha_d x^d + \cdots$ and $c(x) = \gamma_d x^d + \cdots$, the weight satisfies $w(x) \to -\gamma_d/\alpha_d$ as $x \to \infty$.

4.2. Jacobi matrix connection.

The Wallis recurrence (4) defines a semi-infinite Jacobi matrix

$$J = \begin{pmatrix} b(0) & 1 & & \\ 1 & b(1) & 1 & \\ & 1 & b(2) & \ddots \\ & & \ddots & \ddots \end{pmatrix}.$$

The characteristic polynomials $p_n(t)$ of the $n\times n$ truncations satisfy $p_{n+1}(t) = (t-b(n))\, p_n(t) - p_{n-1}(t)$ and are orthogonal with respect to the spectral measure μ of J.

The PCF limit is

$$K = \langle e_0, J^{-1} e_0 \rangle = \int \frac{d\mu(t)}{t}. \tag{8}$$

Remark 4.1. The spectral measure μ of the Jacobi matrix J and the Sturm–Liouville weight $w(x) = -c(x)/b(x)$ are connected by a semiclassical correspondence: in the large-n limit, the equilibrium measure of the orthogonal polynomials $p_n(t)$ is governed by the potential $V(x) = -\int w(x)\, dx$ associated with the SL operator. The precise formulation of this correspondence, and whether it yields an explicit integral representation of K, is an open problem.

5. Classical Examples: Modified Bessel Function Ratios

The general theory developed in Sections 2–4 applies, in particular, to classical continued fractions for ratios of modified Bessel functions. These provide both a calibration of the theory against known results and a bridge to the special-function literature.

5.1. **The Perron continued fraction.** The classical Perron continued fraction for modified Bessel function ratios (Perron [5], Lorentzen–Waadeland [8], Chapter 4) states:

$$\frac{I_{\nu-1}(z)}{I_\nu(z)} = \frac{2\nu}{z} + \cfrac{1}{\cfrac{2(\nu+1)}{z} + \cfrac{1}{\cfrac{2(\nu+2)}{z} + \cdots}}. \tag{9}$$

This is a PCF with $a(n) = 1$ and $b(n) = 2(\nu + n)/z$ for $n \geq 0$.

Theorem 5.1 (Perron–Bessel identification)**.** *Let $\nu = 1$, $z = 1$. The Perron continued fraction* (9) *becomes the PCF with $a(n) = 1$ and $b(n) = 2(n + 1)$:*

$$\frac{I_0(1)}{I_1(1)} = 2 + \cfrac{1}{4 + \cfrac{1}{6 + \cfrac{1}{8 + \cdots}}} \approx 2.24019372387008974\ldots \tag{10}$$

Moreover:

(i) The continuum ODE is $2(x+1)\,y'' + 2\,y' + c(x)\,y = 0$, which is exact: $\frac{d}{dx}[2(x + 1)\,y'] + c(x)\,y = 0$.
(ii) The unique finite singularity at $s = -1$ (root of $b(x) = 2(x + 1)$) is apparent with indicial exponents $\{0, 0\}$.
(iii) The numerical certificate: PCF convergents at $n = 200$ match $I_0(1)/I_1(1)$ to 55 decimal digits.

Proof. Part (iii) is a numerical verification. Parts (i) and (ii) are immediate from Theorems 2.2 and 3.1 applied to $b(x) = 2(x + 1) = 2x + 2$, which has a unique simple root at $s = -1$. The classical identity (10) is the Perron formula (9) evaluated at $\nu = 1$, $z = 1$. □

Remark 5.2 (Degenerate discriminant). Since $b(x) = 2x+2$ is linear, $\mathrm{disc}(b) = 0$ in the naïve sense (a linear polynomial has no pair of roots to discriminate). The single apparent singularity at $s = -1$ does not generate a non-trivial monodromy group, and the structural exclusion (Theorem 7.1) applies *a fortiori*: the singularity carries no arithmetic information about the limit $I_0(1)/I_1(1)$.

5.2. **The $\nu = 1$, $z = 2$ case.** Setting $\nu = 1$, $z = 2$ in (9) gives $b(n) = 2(\nu + n)/z = n + 1$, and the PCF becomes

$$\frac{I_0(2)}{I_1(2)} = 1 + \cfrac{1}{2 + \cfrac{1}{3 + \cfrac{1}{4 + \cdots}}} \approx 1.43312742672231\ldots \tag{11}$$

with $b(n) = n+1$ (degree 1), apparent singularity at $s = -1$, and $\mathrm{disc}(b) = 0$.

This example is especially illuminating: the partial denominators $b(n) = n + 1$ are the simplest non-constant polynomial, yet the limit $I_0(2)/I_1(2)$ is a transcendental number expressible in terms of Bessel functions. The self-adjoint structure and apparent singularity theorem apply, confirming that the ODE $(x + 1)\, y'' + y' + c(x)\, y = 0$ has trivial monodromy at $x = -1$.

6. Relation to Classical Asymptotic Theory

The passage from a difference equation to a differential equation has a long history. In this section we delineate what is classical and what is new in our results.

6.1. The Poincaré–Perron theorem.

The foundational result on asymptotic behavior of solutions to linear recurrences is the Poincaré–Perron theorem (Poincaré [3]; Perron [4]).

Theorem 6.1 (Poincaré–Perron, simplified form). *Let $y_{n+1} = a_n\, y_n + b_n\, y_{n-1}$ be a second-order linear recurrence with $a_n \to a$ and $b_n \to b$ as $n \to \infty$, where the characteristic equation $z^2 - az - b = 0$ has roots λ_1, λ_2 with $|\lambda_1| > |\lambda_2|$. Then every non-trivial solution either satisfies $y_{n+1}/y_n \to \lambda_1$ or $y_{n+1}/y_n \to \lambda_2$.*

For the Wallis recurrence $Q_{n+1} = b(n)\, Q_n + Q_{n-1}$ with $b(n) = \alpha_d n^d + \cdots$, the coefficients grow polynomially, so the Poincaré–Perron theorem does not apply directly (it requires convergent coefficients). The appropriate generalization is the Perron–Kreuser theorem (Perron [4], Elaydi [11], Chapter 8), which gives $Q_{n+1}/Q_n \sim b(n)$ as $n \to \infty$—a statement about the *growth rate* of solutions.

What Poincaré–Perron does not provide. The Poincaré–Perron theorem and its generalizations address the *asymptotic* ratio Q_{n+1}/Q_n as $n \to \infty$. They say nothing about:

- the *structure* of the continuum ODE (whether it is self-adjoint or exact);
- the *singularity type* of the continuum ODE (apparent vs. genuine);
- the *monodromy* of the continuum ODE around its singularities.

Our Theorems 2.2 and 3.1 are statements about the exact continuum limit, not the large-n asymptotics, and are therefore complementary to, rather than consequences of, the Poincaré–Perron theory.

6.2. The Birkhoff–Trjitzinsky theory.

Birkhoff [6] and Birkhoff–Trjitzinsky [7] developed a systematic asymptotic theory for linear difference equations with polynomial coefficients—the discrete analogue of the WKB method. Their theory provides:

(a) formal asymptotic solutions $Q_n \sim C\, n^\alpha\, r^n\, (1 + c_1/n + c_2/n^2 + \cdots)$ for the Wallis recurrence;
(b) Stokes phenomena: the asymptotic series change form across certain directions in the complex n-plane;
(c) connection formulae relating the asymptotic solutions in different sectors.

The modern treatment is in Wimp [12] and Elaydi [11]. For the Wallis recurrence with polynomial $b(n)$, the Birkhoff–Trjitzinsky expansions give the dominant growth $Q_n \sim C \cdot \prod_{k=0}^{n-1} b(k)$ (when $a(n) = 1$), which is the discrete analogue of the WKB leading term $y(x) \sim A(x)^{-1/2} \exp(\int \sigma\, dx)$.

What Birkhoff–Trjitzinsky does not provide. The BT theory works in the asymptotic regime (large n) and produces formal series that are typically divergent. It does not:

- produce an *exact* differential equation satisfied by the continuum limit (only asymptotic approximations);
- detect the *apparent singularity* structure of the exact ODE—the statement $p_0 = 1$, $q_0 = 0$ at each root of $b(x)$ is invisible to the BT expansion, which does not resolve individual singularities;
- establish the *self-adjoint* (exact) form of the ODE, since the notion of exactness requires the full ODE, not just its asymptotic skeleton.

6.3. **The $n \to x$ passage in special function theory.** The passage from a three-term recurrence to a differential equation is classical for special functions:

- *Bessel functions*: the recurrence $J_{n+1}(z) = (2n/z)\, J_n(z) - J_{n-1}(z)$ corresponds to the Bessel ODE $z^2y'' + zy' + (z^2 - n^2)y = 0$.
- *Legendre polynomials*: the recurrence $(n+1)P_{n+1}(x) = (2n+1)x\, P_n(x) - nP_{n-1}(x)$ corresponds to the Legendre ODE $((1-x^2)y')' + n(n+1)y = 0$.
- *Hypergeometric functions*: contiguous relations give three-term recurrences whose continuum limits are the Gauss hypergeometric ODE.

In each of these classical cases, the recurrence has *non-constant* numerator coefficients $a(n)$ (e.g., $a(n) = -1$ for Bessel, $a(n) = -n/(n+1)$ for Legendre), and the resulting ODEs are *not* generically of the exact form $\frac{d}{dx}[A(x)\, y'] + C(x)\, y = 0$.

The self-adjoint property of the Legendre ODE, for instance, arises from a different mechanism: the weight function $(1-x^2)$ in the Legendre operator is prescribed *a priori* by the orthogonality interval $[-1, 1]$, not by the derivative relation $B(x) = A'(x)$.

Our contribution. For PCFs with $a(n) = 1$ (constant numerator), the passage $n \to x$ *always* produces a self-adjoint ODE—Theorem 2.2. This is a new structural result specific to the constant-numerator case, not a consequence of the Poincaré–Perron theorem, the Birkhoff–Trjitzinsky theory, or the classical $n \to x$ passage for special-function recurrences with variable numerators. The mechanism is the constancy of the discrete Wronskian (Theorem 2.1), which has no analogue in the variable-numerator setting.

6.4. **Summary of novelty.** Table 1 collects the comparison.
The table makes clear that the self-adjoint structure theorem, the apparent singularity theorem, and the discriminant universality result are genuinely new: they are not covered by any of the three classical frameworks. The key innovation is the identification of the *constant discrete Wronskian* as the mechanism linking the algebraic structure of the recurrence ($a(n) = 1$) to the analytic structure of the continuum ODE ($B(x) = A'(x)$).

TABLE 1. Comparison of the present results with classical theories.

	PP theorem	**BT theory**	**Classical** $n \to x$	**This paper**
Asymptotic ratio Q_{n+1}/Q_n	✓	✓	✓	—
WKB-type expansion	—	✓	—	—
Stokes phenomena (discrete)	—	✓	—	—
Exact continuum ODE	—	—	✓	✓
Self-adjoint structure	—	—	*case-by-case*	**general**
Apparent singularity theorem	—	—	—	✓
Discriminant universality	—	—	—	✓
Trivial monodromy	—	—	—	✓
Arithmetic exclusion	—	—	—	✓

7. Consequences for Arithmetic Identification

Theorems 3.1 and 3.2 together explain a persistent phenomenon in the PCF arithmetic classification program: PSLQ-based searches for PCF limits against CM-period bases at discriminant $\mathrm{disc}(b)$ return null results.

Corollary 7.1 (Structural exclusion)**.** *Let $K = b(0) + K_{n\geq 1}\, 1/b(n)$ with $\mathrm{disc}(b) = \Delta$. Then K cannot be expressed as a CM period, gamma value, or hypergeometric special value whose monodromy involves the singularities $\{s_k\}$ of the associated ODE, because the monodromy at each s_k is trivial.*

In particular, the "CM discriminant" $\Delta = \mathrm{disc}(b)$ is a property of the Wallis characteristic polynomial, not an arithmetic invariant of K. Searches against CM bases at discriminant Δ fail a priori*, not merely at finite precision.*

This corollary generalizes Theorem 3.1 of [1] from the specific case $b(x) = 3x^2 + x + 1$ ($\Delta = -11$) to arbitrary polynomials. It explains, for example, why the $d = 3$ PCF with $b(n) = n^3 + n + 1$ ($\Delta = -31$) would likewise resist identification via Heegner-point methods or Stark-class-number approaches at discriminant -31.

8. Examples

8.1. **Degree 1: $b(n) = 2n + 1$ and the coth identity.** The simplest non-trivial case. The Wallis recurrence is $Q_{n+1} = (2n+1)\, Q_n + Q_{n-1}$. The ODE:

$$(2x+1)\, y'' + 2\, y' + c(x)\, y = 0.$$

Single finite singularity at $s = -1/2$ (apparent, $\rho = 0, 0$). Discriminant: $\mathrm{disc}(2x+1) = 1$ (trivial).

The PCF value is

$$K \;=\; 1 + \cfrac{1}{3 + \cfrac{1}{5 + \cfrac{1}{7 + \cdots}}} \;=\; \coth(1) \;=\; \frac{e^2+1}{e^2-1} \;\approx\; 1.31303528549933\ldots$$

This is the classical Stieltjes S-fraction for the hyperbolic cotangent (Lorentzen–Waadeland [8], §4.2). The half-integer modified Bessel functions $I_{-1/2}(z) = \sqrt{2/(\pi z)}\cosh z$ and $I_{1/2}(z) = \sqrt{2/(\pi z)}\sinh z$ give the identification

$$K \;=\; \coth(1) \;=\; \frac{I_{-1/2}(1)}{I_{1/2}(1)}.$$

(Note: the integer-order ratio $I_0(1)/I_1(1) \approx 2.2402$ corresponds to the distinct PCF with $b(n) = 2n + 2$.) The Jacobi matrix J with diagonal entries $b(n) = 2n+1$ and unit off-diagonal has a discrete spectral measure supported on the zeros of certain Bessel functions, rather than a continuous weight.

Remark 8.1 (Stieltjes weight refutation). The Sturm–Liouville weight for this example is $w(x) = -c(x)/b(x)$. A natural conjecture is that the PCF limit admits a Stieltjes-type integral representation $K = \int w(x)\, g(x)\, dx$ for some kernel g. Numerical experiments at 50 decimal digits refute this: for three test cases ($b(n) = 2n + 1$, $3n^2 + n + 1$, $n + 1$), seven distinct integral kernels (including 1, $1/x$, $1/(1+x)$, $1/(x(1+x))$, $\log x$) yield values differing from K by at least 0.4, and PSLQ detects no integer relation among the integral values and K. The obstruction is immediate: $w(x) \to \text{const}$ as $x \to \infty$ for $\deg b \geq 2$, so w is not integrable on $[0, \infty)$.

The correct integral representation uses the spectral measure μ_J of the Jacobi matrix rather than the SL weight; see Theorem 9.2.

8.2. **Degree 2: V_{quad} (worked example).** $b(n) = 3n^2+n+1$, $c(x) = -x^2$. Two complex-conjugate singularities at $s_{1,2} = (-1\pm i\sqrt{11})/6$. Both apparent. $\text{disc}(b) = -11$. $K = V_{\text{quad}} \approx 1.94264768946523$.

This is the motivating example: the identity $b(x) \equiv A'(x)$ was first observed for V_{quad} in [1], where it was used to prove the apparent singularity exclusion theorem. The general theory of the present paper shows that this identity is automatic for any PCF with $a(n) = 1$, regardless of the degree of $b(n)$.

Unlike the Bessel-function examples of Section 5, the limit V_{quad} does not match any entry in the Inverse Symbolic Calculator or PSLQ searches against known constants at 600-digit precision [1].

8.3. **Degree 3: $b(n) = n^3 + n + 1$.** Three finite singularities: one real ($s_1 \approx -0.6823$) and a complex-conjugate pair ($s_{2,3} \approx 0.3412 \pm 1.1615\, i$). All three are apparent singularities with $p_0 = 1$, $q_0 = 0$, verified both symbolically (by Theorem 3.1) and numerically (Frobenius computation at 150-digit precision). $\text{disc}(b) = -31$. The PCF converges rapidly:

$$K = 1 + K_{n\geq 1}\, \frac{1}{n^3 + n + 1} \approx 1.32355722450018.$$

This is the first degree-3 example verified under the general theory. The discriminant -31 is a Heegner discriminant (class number 3), but the structural exclusion (Theorem 7.1) implies that this coincidence carries no arithmetic content: $\text{disc}(b)$ is a property of the polynomial $b(x) = x^3 + x + 1$, not of the limit K.

9. Open Problems

Open Problem 9.1. For which non-constant polynomials $a(n)$ does the continuum ODE still satisfy $B(x) = A'(x)$? The condition $a(n) = \text{const}$ is sufficient; is it necessary among polynomials?

Open Problem 9.2 (Spectral measure and integral representation). Let J be the Jacobi matrix with diagonal entries $b(n)$ and off-diagonal entries 1. The spectral measure μ_J of J is a probability measure on $\mathbb{R}$, and the PCF limit satisfies

$$K = \int \frac{d\mu_J(t)}{t}$$

(the Stieltjes transform of μ_J at $z = 0$, assuming $0 \notin \operatorname{supp}(\mu_J)$). The Sturm–Liouville weight $w(x) = -c(x)/b(x)$ determines the Jacobi matrix asymptotically via the inverse spectral problem, but does not yield a direct integral formula: numerical experiments confirm that $\int w(x)/(x(1+x))\,dx$ and six related kernels do not match K for $b(n) = 3n^2 + n + 1$ (see Theorem 8.1).

The calibration case $b(n) = 2n+1$ gives $K = \coth(1)$ exactly, with spectral measure supported on zeros of Bessel functions rather than on the continuous weight $1/(2x+1)$. Computing μ_J explicitly from $b(n)$ via the inverse spectral problem, and evaluating $\int t^{-1}\,d\mu_J(t)$ in closed form, is the central open problem of this theory.

Open Problem 9.3. Classify the Sturm–Liouville weight functions $w(x) = -c(x)/b(x)$ arising from PCFs. When is $w(x) \geq 0$ on $[0, \infty)$?

Open Problem 9.4. For PCFs with multiple roots of $b(x)$ (i.e., $\operatorname{disc}(b) = 0$): the corresponding ODE singularity becomes irregular. What replaces the apparent singularity theorem in this degenerate case?

Open Problem 9.5 (Characterization of the self-adjoint class). The condition $a(n) = \text{const}$ is sufficient for $B(x) = A'(x)$ (Theorem 2.2). Is there a wider class of pairs $(a(n), b(n))$ with $a(n)$ non-constant for which the self-adjoint structure persists? For example, does $a(n) = (-1)^n$ (alternating sign but constant absolute value) preserve the property? A complete characterization of all (a, b) yielding exact continuum ODEs would delineate the boundary of the self-adjoint PCF theory.

Open Problem 9.6 (Inverse discriminant problem). For degree-d PCFs with $a(n) = 1$ and $b(n)$ of degree d, the discriminant $\operatorname{disc}(b) \in \mathbb{Z}$ depends only on the coefficients of b. Which integers arise as $\operatorname{disc}(b)$ for some polynomial $b \in \mathbb{Z}[x]$ of degree d? For $d = 2$, the answer is: all integers of the form $\beta^2 - 4\alpha\gamma$ with $\alpha, \gamma \geq 1$ and $\beta \in \mathbb{Z}$, which includes all integers $\equiv 0$ or $1 \pmod 4$. For $d \geq 3$, the set of realizable discriminants is less understood and connects to the classical theory of algebraic integers and the inverse Galois problem.

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